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Prediction and diagnosis of cardiovascular disease using cloud and machine learning design

K. Babu[1], A. Gokula Chandar[2] and S. Kannadhasan[3*]

Abstract
Predicting and accurately identifying heart disease is a significant challenge in the field of medicine, and the problem of cardiovascular disease predetermine in the health care system is regarded as an essential challenge. Patients have access to more expensive surgical procedures at these rapidly expanding health care organisations. Recent years have seen an increase in the prevalence of heart disease; this means that despite the progress that has been made in medicine, the prevalence of cardiovascular disease continues to rise at an alarming rate. The primary contributors to the development of these illnesses are a sedentary lifestyle, excessive use of alcohol, insufficient time spent being physically active, and the use of cigarette products. As a result, there is a requirement for a cloud-based framework (CBF) that is capable of monitoring health information and making accurate predictions regarding it. Recently, techniques from the field of machine learning have been applied in an effort to address issues of this nature. But the method that is being suggested uses a cloud-based and cloud-based four-step process to improve surveillance of patients' health information. This is done to improve the process of forecasting patients' health information. Detecting and categorising cardiac illness can be accomplished through the application of two distinct kinds of machine learning techniques. After that, an analysis is performed to determine how accurate those techniques are. In order to assess how effectively they work, evaluation parameters are utilised.

Keywords Heart disease, Prediction, Classification, CBF, Machine Learning Algorithms (MLA)

Introduction

Nowadays people's lifestyle is increasing day by day due to inheritance. This creates not only time but also a lot of data, thus wasting data on health surveys.But nowadays data analysis has come into play.The project was developed to create a large amount of data from hospitals to be effective. Heart disease is a deadly enemy during this period.The disease can be cured as much as possible otherwise it affects that particular person. And the right time is to diagnose patients, and it is a very difficult job. Hospital tests can sometimes be misdiagnosed, leading to bad names.The disease is very difficult to treat and most patients are unable to treat it.

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In the paper [1], two algorithms in machine learning have been used to predict heart disease. In paper [2], machine learning is used to predict the presence of heart disease.In paper [3], a hybrid machine learning method has been used to diagnose heart disease. In paper [4], data mining technique has been used to diagnose and monitor heart disease. In paper [5], the MLA system has been used to diagnose and predict atherosclerosis. The process of data mining incorporates a significant component that is known as categorization. The process of establishing a model begins with the classification of the data, which involves recognising the various kinds of information or concepts that are being represented. An extensive amount of data elements are analysed in order to arrive at a conclusion regarding the model. accumulating statistics on cardiovascular conditions (also known as data elements for which the The grades have been disclosed). It is not at all obvious whether the algorithm is used to predict the outcomes of the class name of the objects for which the class name is having the cardiovascular disease or not having cardiovascular sickness. That is not at all clear. The field of study known as machine learning investigates how computers pick up new skills. can learn from (or put to use to improve their performance) in light of information regarding their cardiopulmonary system. Computer efforts have been designated as the primary research area in order to eventually learn how to understand complex situations and ultimately choose resourceful options based on cardiovascular statistics. The goal of this research is to be accomplished through the use of computers [6]. The analogous concept of organisation is an important part of controlled learning and should not be overlooked.

As a result of this, it is absolutely necessary to keep track of as many particulars as is humanly feasible. both favourable and negative inclinations in terms of health that contribute to the development of cardiovascular disease. Several studies are performed in order to gather information prior to making a decision. A couple of instances of CVD diagnostic techniques include auscultation, ECG, cardiovascular exercise, and others. Cholesterol and hyperglycemia are also included. They are capable of machine learning, which was utilised seven times, and each of the three components was understood. computations based on decision, a method known as cross-approval, and seven classifications performance evaluation measurements. These measurements include things like sharpness in characterisation, particularity in expression, affectability, Matthews' relationship coefficient, as well as the amount of time it took to carry out. They have created a gadget that is undeniably capable of distinguishing between healthy people and those who suffer from cardiovascular disease in order to save people's lives. They have discussed the general categorization methods, feature sets, and computations for assurance, preliminary steps, and official clearance methodology, as well as the classifiers performance assessment approximations that were utilised in this specific research. They have completed the steps necessary to put the proposition into action. Device has been recommended due to both the comprehensive set of features that it possesses as well as the decreased cost that it offers. the arrangement of the components of the game. Because of who they are and what they are, influences have less of an impact on them. The time has arrived to implement classifications with regard to the exactness of their application and the manner in which they are carried out. They recommended learning strategies based on the use of computers. The professionals will find that choice-based networks that are emotionally compassionate will be of assistance to them in their efforts to successfully recognise individuals who have cardiac conditions. S. The future can be accurately predicted using a technique discovered by Krishnan. putting someone at risk of developing a cardiac condition. Because of the aftereffects of this procedure, the risk of something happening to the heart is increased [4, 7–10].

They have taken into consideration the fact that the databases that were used are organised in a fashion that is comparable to that of the therapeutic parameters. Their methodology makes use of the aforementioned information in order to carry out an analysis of the aforementioned characteristics. methodical strategy for the separation process. Python was utilised throughout the process of their libraries being prepared. using two approaches that are considered industry standards for machine learning in your programming. It has been demonstrated that the decision tree algorithm, the naive Bayes Algorithm, and both of these together are the most accurate projection techniques. This is necessary for the algorithm to be obvious. Among these two, pre-planning makes use of a variety of techniques, such as the elimination of noisy data, the elimination of insufficient data, and the filling in of benchmark values if the data are missing. This is because pre-planning is used to determine the degree of precision necessary for the information regarding cardiac disease. classification of characteristics that are applicable and helpful for the process of decision making on numerous levels of complexity. The demonstration that they provided of the Obtaining a model is accomplished through the application of techniques such as order, investigation of exactness, affectability, and particularity. This is a forecasting algorithm that was developed to determine whether or not individuals have coronary illness, whether or not they exercise mindfulness, and whether or not they investigate on the subject. They have carried out an investigation by comparing the levels of accuracy demonstrated in the process of putting

in place guidelines with regard to the specific effects of Support Vector Machine, Gradient Boosting, Random wilderness, Random wilderness, The information was analysed using a Naive Bayes classifier, and recurrence was determined. accumulated within the confines of a jurisdiction for the purpose of presenting an accurate model of anticipating cardiovascular illness. Cardiovascular illness, and they have taken into account both the masculine and feminine classes, and this percentage may differ depending on the geographic region, and additionally, this percentage of individuals between the years of 25 and is taken into consideration. 69. This does not demonstrate that individuals in different age groups are fundamentally distinct from one another. The presence of coronary conditions will have no impact whatsoever on the gathering. They are able to anticipate the reason, and at this point, cardiovascular illness functions as an important indicator. They've gone over a variety of calculations, which are both diagnostic and prognostic tools for cardiac illnesses, and they've talked about how they work. They were able to accomplish this through the utilisation of the established regression technique, and pharmacological services were offered. according to the information in the document, whether the individual had cardiac conditions or not, and developed a statistical algorithm that can determine whether or not a subject has are suffering from a cardiac condition or not. information that organises the individuals according to whether or not they are according to whether or not they are suffering from a cardiac condition or not.

Cardiovascular disease, and made an inappropriate use of the Fast Correlation-Based Strategy also known as Feature Selection (FCBF) in order to assist in redirecting overflow concentrates on in order to acquire a better understanding of the characteristics of cardiovascular disease order [11–15]. They have arranged things based on various organisation-related computation methods, such as K-Nearest Neighbor, the Support Vector Machine, the Naive Bayes Classification, the Random Forest Classification, and a Multilayered Perception Simplified Artificial Neural Network by utilising Particle Swarm Optimization (PSO) in conjunction with Approaches using the Colony Optimization (ACO) algorithm. In addition, they have created a Multilayered Perception Simplified Artificial Neural Network. It was their idea in the first place. Using composite methodology, an analysis of the data on cardiovascular illness was carried out. Using the enhanced model that was recommended by the FCBF, PSO, and ACO, we were able to accomplish the greatest degree of precision possible, which was 99.65%. Cardiac illness is a condition that poses a significant risk to one's health and can ultimately result in cardiac failure. Heart disease is a significant condition that has a negative impact on the function of the heart and can lead to complications such as impaired blood vessel function and inflammation of the coronary arteries. Heart illness, also referred to as a cardiovascular disease, is one of the leading reasons of mortality on a global scale. The muscular structure that is responsible for pumping blood throughout the body is called the heart. Heart illness is an umbrella term that refers to a number of conditions that impact the cardiovascular system, specifically the heart and blood vessels. Heart illness can take many forms, some of which include coronary artery disease, angina, a heart attack, and heart failure. Coronary heart disease, also known as CHD, is one of the primary factors contributing to sickness and death in our modern civilization.

Because of the substantial financial weight that is incurred when treating coronary artery disease, prevention of coronary artery disease is an essential stage in the treatment process. When a coronary artery abruptly becomes obstructed, the result is a heart attack. The blockage is typically caused by a blood clot. High blood pressure, coronary artery disease, cardiac valve disease, and stroke are the different kinds of cardiovascular diseases that can affect a person's health. According to the World Health Organization, cardiovascular illness was responsible for the deaths of 17.7 million people. It is of the utmost importance that individuals figure out how to comprehend and control the unintentional variables that contribute to cardiovascular disease. Some of these factors include a healthy diet, regular physical activity, and medication prescribed by a physician to treat blood pressure, cholesterol, and weight. A few fatalities can be averted by taking preventative measures against coronary disease at an early stage. The identification of cardiac illness is complicated by the presence of a number of different variables that need to be considered. Age, gender, blood sugar while fasting, chest discomfort type, and the ability to relax are the fundamental characteristics used to diagnose heart disease. Electrocardiogram, massive fluoroscopic shad arteries, tests for high blood pressure (hypertension), serum cholesterol (coronary artery disease), and other diagnostic procedures disease risk factor), thalach (greatest pulse), ST melancholy, fasting hyperglycemia, exang (including angina), tobacco, and alcohol consumption. hypertension, a preference for certain foods, one's weight, their height, and tightness. Pain in the chest and arms, sluggishness and Some of the early warning indications of a heart attack include feeling lightheaded, exhausted, and sweating excessively. For the process of identifying conditions related to the heart, data mining and ensemble approaches are utilized, whose findings are most trustworthy as well as accurate. As a result of a variety of risk factors, including cholesterol, diabetes, and elevated blood pressure. It can be difficult

to diagnose the illness in individuals due to stress as well as a number of other variables [16–20].

In addition, techniques based on machine learning are taken into consideration for the projection illness caused by the Corona virus. According to the American Health Association, considerable amounts of money are spent on health treatment. It is projected that the number of deaths caused by cardiovascular illness will increase by the year 2030. Every nation has a greater number of institutions and increased numbers of patient information as a direct result of population development. The wellbeing of their patients is prioritised at most institutions. Data mining, also known as information discovery in databases, is so named because it uses a number of different techniques. strategies utilised in a variety of industries, including but not limited to machine learning, neural networks, and information extraction. Data mining is important to uncovering concealed patterns and developing analytical frameworks in order to successfully retrieve information. The approach known as application analysis should be able to recognise patterns operating at multiple layers of abstraction. a variety of data gathering techniques, including machine learning algorithms. It is anticipated that will have a greater degree of precision. In light of this, the following are some strategies and potential selection methods that will assist the accuracy have been achieved in the healthcare business. The data extraction strategies that are currently being implemented in the medical industry. There are seven different techniques for machine learning.

The primary objective of this study was to establish a reliable algorithm for predicting cardiovascular illness in Jammu and Kashmir. (India). In order to accomplish this goal, machine learning techniques combined with IoT were utilised. The model for making predictions is constructed using the characteristics. The specifications are decided upon following discussions with the specialists in the relevant field. The findings of this study led the researchers to the conclusion that the Naive Bayes algorithm performed better when it was not subjected to hyper tuning of its parameters, whereas the Random Forest model was shown to be an effective method even when its parameters were subjected to hyper tuning. Because the Hyperparameterized Random Forest model has better precision and lower mistake rates in comparison to all of the other approved models, we have decided to make the ultimate result using the Hyperparameterized Random Forest model. In the years to come, the continuation of this line of research will focus on making predictions regarding the outcomes of elections across the entirety of India. This model can only forecast cardiovascular illness for Jammu and Kashmir at the moment, but researchers plan to use this methodology to expand their research work to other developing areas by incorporating certain characteristics that are specific to those areas. The new algorithm could be developed by additional researchers, or an ensemble methodology could be used to integrate all of the algorithms in order to accomplish a higher level of precision with a lower rate of mistake. In today's world, the healthcare industry generates an enormous quantity of data relating to the identification of diseases, patients, and other related topics. The process of "data mining" involves making use of a variety of methods in order to unearth concealed patterns or commonalities within data. For this reason, the purpose of this article is to suggest a machine learning algorithm for the implementation of a heart disease prediction system, which was then confirmed using two open access heart disease prediction datasets. The introduction of a cardiac patient monitoring system that uses the idea of the Internet of Things (IoT) with various physiological signal devices and an Arduino microcontroller is another one of this paper's contributions. The technology known as the Internet of Things (IoT) is presently being implemented into sensor networks in order to facilitate the gathering, analysis, and transmission of information from one point to another. The Internet of Things (IoT) is a relatively new and quickly developing technology that allows numerous devices and data aggregators to perceive their surroundings, exchange information with one another, and communicate using either a private network, Internet Protocol (IP), or public networks. After a predetermined amount of time, the devices capture the data, process it, and use it to instigate the necessary action. Additionally, they provide a sophisticated cloud-based network for the purposes of research, planning, and decision making. It was estimated that between 8 and 50 billion devices will be connected to the internet by the year 2020. The products that are developed with Internet of Things, such as integrated technology, enable for the interchange of information among other networks or the internet.

The purpose of this study is to develop a Decision Support System for the diagnosis of cardiac illness. The system will use a data mining methodology that has the highest precision and performance among Naive Bayes, Support Vector Machine, Simple Logistic Regression, Random Forest, and Artificial Neural Network (ANN), among others. It is possible to evaluate the likelihood of developing heart disease by taking several cardiovascular system characteristics into consideration, including age, blood pressure, the findings of an electrocardiogram (ECG), gender, and blood sugar levels. An in-depth comparison of several distinct machine learning algorithms has been presented as a means of determining which of these approaches has the potential to provide the most accurate diagnosis and prognosis of cardiac illness. As inputs, the medical characteristics such as age, blood

pressure, heartbeat, sex, ECG findings, blood sugar, and so on are all taken into consideration by this algorithm, which then calculates the likelihood of an individual developing heart disease as an output. The concept and design of an android application that is web-based and makes use of an effective machine learning strategy are included in this suggested system. This application is intended to diagnose cardiac illness. When it comes to diagnosing cardiac illness, it can be a very helpful instrument for both patients and physicians, as well as students of medicine. After being discharged from the hospital, a patient's health must be monitored continuously for a full day and night before a determination of potentially deadly physiological conditions and symptoms, such as a heart attack, can be made. Using this application, the patient can view the danger level of acquiring heart disease and enter the present characteristics of their heart disease from anywhere on the application dashboard. All of the parameters that are not measured in real time, such as blood sugar, serum cholesterol, and ECG results, will be included in the report that the doctor prescribes for the patient. However, there are some parameters, such as chest pain type and exercise-induced angina, that the patient is responsible for self-measuring on a periodic basis and inputting the values manually on the application interface [21–24]. In the event that the application identifies a potentially deadly circumstance, the patient has the ability to communicate with any physician via video chat, and any authorised physician who specialises in heart disease can be located by entering his or her phone number into the search box.

Proposed methodology

Data-set

The final attributes are selected first for pre-processing. This data set contains the information of 300 people. And of these there are fifteen columns. It is explained below.

1. ***Age***:

The age of each person is indicated here.

2. ***Sex***:

Of these, gender is defined using the forms given below.

1 = female
0 = male

3. ***Chest-pain type (CPT)***:

It describes the chest pain that people experience using the following guidelines

1 = asymptotic
2 = non-angina pain
3 = atypical angina
4 = typical angina

4. ***Resting blood pressure (RBP)***:

The unit of mm-Hg is shows the RBP value of an individual.

5. ***Serum cholesterol (SC)***:

The unit mg/dl shows SC.

6. ***Fasting blood sugar (FBS)***: The FBS value of a human is compared to that of 120 mg/dl.

if
FBS > 120 mg/dl
then:1(true)
else
0 (false)

7. ***Resting ECG (RECG)***: This shows RECG results,

0 means normal
1 means it having ST-T wave abnormality
2 means left ventricular hypertrophy

8. ***Max heart rate (MHR)***: It shows the MHR reached by a human.

9. ***Exercise induced angina (EIA)***:

1 means yes
0 means no

10. ***ST depression (STD)***: It shows the value that is a float or an integer (Table 1).

Architecture

In this proposed method, machine learning systems, including cloud, are designed here. Here are four steps designed to monitor and predict the incidence of heart disease. Devices and monitoring sensors are used to monitor people's health. So, in step 1, that surveillance collects data from sensors and devices and combines them.In the second step, as the data is monitored and collected daily, the data is getting bigger. A huge amount of device is needed to collect all this. It uses the cloud server to store large an amount of real-time information. Step 3 is developing a real-time based design for early detection of heart disease. It uses the MLA necessary for prediction and classification to create them. Step 4 sends

Table 1 Data-set

Index	Age	Sex	CPT	RBP	SC	FBS	RECG	MHR	EIA	STD
1	64	0	2	140	234	0	0	151	1	2.4
2	68	0	3	165	287	1	0	109	0	1.6
3	68	0	3	125	230	1	0	130	0	2.7
4	38	0	4	120	251	1	2	188	1	3.
5	42	1	1	120	205	1	0	173	1	1.5
6	57	0	1	110	237	1	2	179	1	0.9
7	63	1	3	150	269	1	0	161	1	3.7
8	58	1	3	130	355	1	2	164	0	0.7
9	64	0	3	140	255	1	0	148	1	1.5
10	64	0	3	130	204	0	0	156	0	3.2

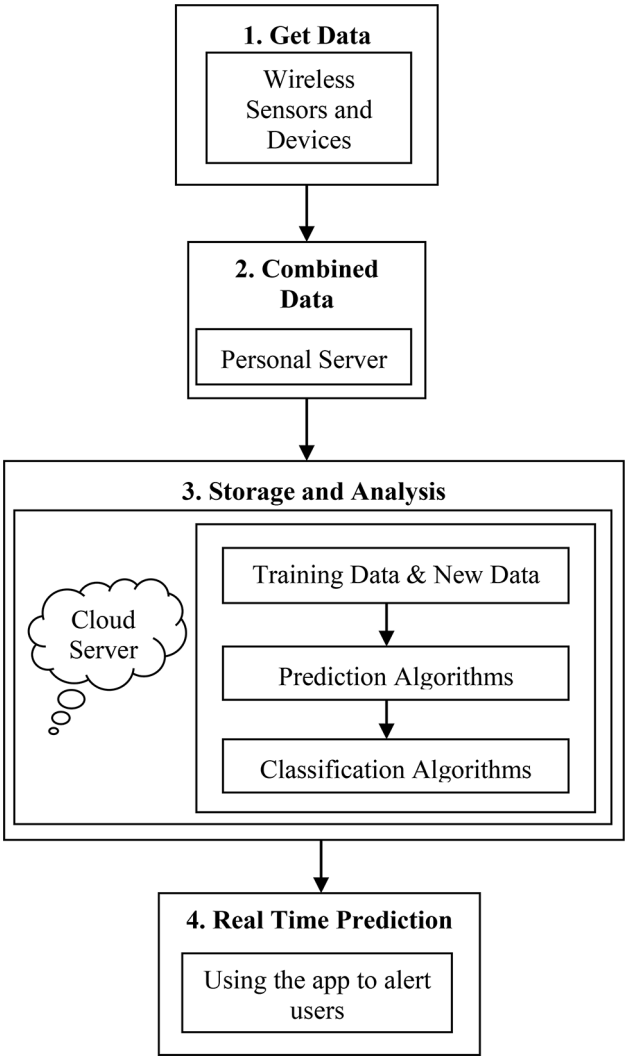


Fig. 1 Proposed architecture

the end result to the users. Here is using an application to do so.The structure of these four steps is given in Fig. 1.

Requests from the monitoring system are accepted and stored by the cloud server. If the new data is equal to pre trained data, that data is stored in cloud server. If the new data is or higher than or lower than or not equal to the pre trained data, that data go to as notification to user. In this paper, for heart disease, to come up with a better solution, provides MLA, cloud servers, NoSQL database and real-time predictive service.

The main contributions of this paper are mentioned in the following points:

- Using machine learning, cloud-based systems are recommended for monitoring and predicting heart disease. This system helps to cardiologists make good, effective decisions.
- From various health monitoring services, to handle big data like real-time data and health services data, this paper deals with many excellent systems. If this paper does not handle this large amount of information, some of the important features of that data will be missed.
- Most analyzes do not use real-time settings. But this paper just uses it. And some studies are based on accuracy only. But this paper is all about accuracy and real-time system (Fig. 2).

There are four steps in this proposed method, they are, respectively,

1. Data collection
2. Data storage
3. Analysis module
4. Application presentation

The data collection step is used to extract data from a specific patient using devices and sensors. To continuously collect the health data of a particular person, health monitoring devices are integrated with the human body. Moreover, monitoring systems are sending data seamlessly. To this vast amount of data, by using traditional database techniques and tools, making their analysis and storage difficult.This proposed system uses NoSQL data base and cloud computing technologies to continuously collect health data. And in the app step, users can view their usage reports using the mobile app.

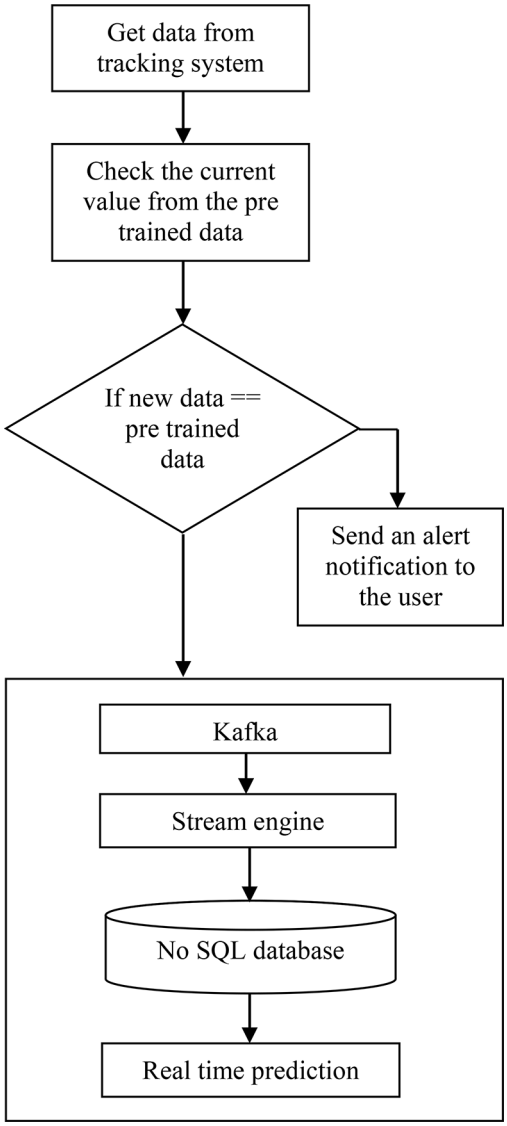


Fig. 2 Flow diagram

Prediction method using K-Nearest neighbors (K-NN) algorithm

The KNN system is a supervised machine learning system. It is used to solve classification and regression problems. The KNN mechanism is based on actual observations in nearest.

Let D_S = {(a_1, b1), (a_2, b_2) ... (a_N, b_N)} Atraining data set (observations set) of q-dimensional patterns with A = $[a]_{j=1}^{N} \subset D^q$, B = $[b]_{j=1}^{N} \subset D^q$ as a corresponding label set, and N is the no. of training entities. In the data space, Minkowski metric is used to define a homogeneity function.

$$\|a' - a_i\|^p = \left\{ \sum_{j=1}^{q} \left| (a_j)' - (a_j)_i \right|^p \right\}^{\frac{1}{p}} \quad (1)$$

Pretty much the distance is said to be the Euclidean distance. This is used to determine the distance between the query and the data set points.In multi-class classification mode, to unknown Model a0, in the data space, the KNN system predicts the majority class label near K.

$$F_{KNN(X')} = \arg \max_{b \in N_k(a')} \sum_{i \in N_k(a')} L(b_j = b) \quad (2)$$

Where, the L() is indicator function. L (b_j = b) = If argument is true, the value is 1 and otherwise 0.The choice of the K value will always be odd numbers.

Classification method using K-means clustering algorithm

Hard classification called KMC consists on a partitioning N training dataset or observations dataset A of n attribute vectors into C classes. The purpose of the classification algorithm is to find class centroids that reduce objective function. The objective function is given by the equation given below:

$$I = \sum_{j=1}^{C} \left\{ \sum_{k, x_k \in v_j} \|a_k - v_j\|^2 \right\} \quad (3)$$

Where, v_j is the centroid of the jth class, d (a_k, v_j) is the distancebetween j^{th}centroid C_j and the kth data of A. Generally, theEuclidean distance is utilized to define the objective process.

Here the data must belong to only one denominator. Therefore, the membership team M has two characteristics. Their equation is given below.

$$\sum_{j=1}^{c} a_{jk} \quad (4)$$

Where, ∀k = 1, ..., N

$$\sum_{j=1}^{C} \sum_{K-1}^{N} a_{jk} = N \quad (5)$$

The value v_j of every and each class centroid is calculated by the mean of all its attribute vectors:

$$v_j = \frac{1}{|C_j|} \sum_{k, a_k \in v_j} a_k \quad (6)$$

Where, $|C_j|$ = the cardinal or the size of C_j.

Results and discussion

Several methods are used for the assessment of atherosclerosis. All these methods are sensitivity (S_E). It quantifies the amount of patients who are most accurately identified as having the disease.

Table 2 CM for binary classification

Predicted output (PO)	Actual outputs (AO)	
	Patient has not disease	*Patient has disease*
Negative Test (NT)	TN	FN
Positive Test (PT)	FP	TP

Table 3 CMof two MLA

	FN	TN	FP	TP
KNN	6	43	5	29
KMC	9	26	22	26

The specificity (S_P) is a computation of the number of patients correctly identified as having no disease. Accuracy (A) depends on the operation of the algorithms used. MCC (Matthews's correlation coefficient) is a measure of the strength used in binary classifications machine learning.

In machine learning, the CM is also called the error matrix. Refer to Table 2 to see the effectiveness of the algorithms.This matrix has 2 information types. That is the efficiency of the predicted algorithm and the actual performance of the algorithms.

The overall efficiencies were presented using equations. The TP (True Positive) is that the patient correctly diagnosed a condition called atherosclerosis. The FP (False positive) is a misdiagnosed prediction of disease.The TN (True Negative) is the prediction that the patient is healthy and fairly accurate. The FN (False negative) is a misdiagnosed prediction that the patient is healthy.

$$S_p = \frac{TN}{TN + FN} \tag{7}$$

$$S_E = \frac{TP}{FN + TP} \tag{8}$$

$$A = \frac{FN + TP}{FP + TP + TN + FN} \tag{9}$$

$$MCC = \frac{TN.TP - FN.FP}{\sqrt{(FP + TP) . (FN + TP) . (FP + TN) . (FN + TN)}} \tag{10}$$

The Clinical Diagnostic Support System for atherosclerosis diagnosis (AD) is provided using the KNN and KMC Methods. There are two types of AD classes. One is healthy and the other is atherosclerosis.There are 280 samples in the NoSQL database.Seventy of percentage is used in training data.And 15% of validation data were used.The remaining data have been used as test data.Two types of machine learning have been used for atherosclerosis.The results of the prediction and classification of the proposed system are built using the Confusion Matrix (CM) in Table 3. Each CM cell contains assorted source number of actual and predicted outputs.

Table 4 Performance of MLA

	KNN	KMC
MCC (%)	0.73	0.27
A (%)	87	62
S_E (%)	84	74
S_P (%)	91	53

The performance results and evaluation of the AD presented in Table 4. The results have been detectedutilizing the KNN, and KMC algorithms. The performance results gave that the proposed system had a sensitivity of 98%, an accuracy of 97%, and specificity of 97% as the good rates.

On the other hand, checking the MCC would be the proposed system performance indicator. In this process, the value of MCC is 0.94 utilizing Eq. (10). If the result is one, the proposed system has a good and correct prediction. The accuracy of the proposed system is arranged in the order of increasing the rate.

Conclusion

In this paper, two supervised learning-based machine learning techniques are described. Then the efficacy of the two classifiers used in the prediction of heart disease was compared. Then their performance is estimated using ten times the cross-sectional and confusion matrices.This paper used machine learning-based cloud architecture to predict and diagnose heart disease. Therefore, a real-time, four-step cloud system has been proposed to monitor and predict heart disease.This test used two machine learning techniques to predict heart disease based on several parameters.In addition, these cloud-based applications were able to detect heart disease by collecting data from the health center.This includes using the cloud system to detect cardiovascular diseases and send warning messages to patients. This system can therefore be used to predict and monitor heart disease.

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K.Babu – Conceptualizaion, Methodology, Implementation and ResultsG, Gokula Chandar - Conceptualizaion, Methodology, Implementation and Results S.Kannadhasan - Conceptualizaion, Methodology, Implementation and Results.

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Declarations

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Competing interests
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